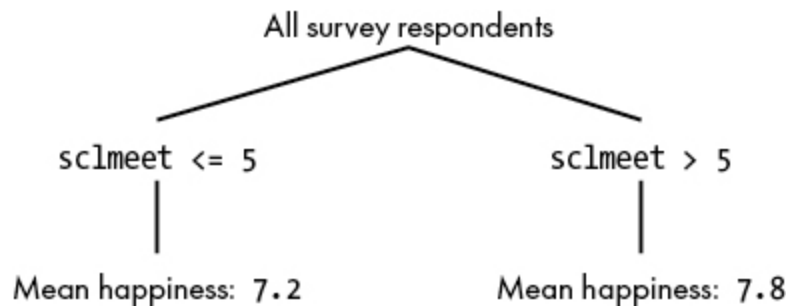


of all people with lower ratings of social activity (which we saved in the variable `low_social_happiness`) and the happiness levels of all people with higher ratings of social activity (which we saved in the variable `high_social_happiness`). Finally, we calculated the mean happiness rating of unsocial people (`meanlower`) and the mean happiness rating of highly social people (`meanhigher`). If you run `print(meanlower)` and `print(meanhigher)`, you should see that people who rated themselves as highly social also rated themselves as slightly happier than their less socially active peers: about 7.8 was the mean happiness level reported by the socially active, and about 7.2 was the mean happiness level for the socially inactive.

We can draw a simple diagram of what we just did, as in [Figure 9-2](#).



[Figure 9-2](#): A simple decision tree predicting happiness based on frequency of social outings

This diagram of our simple binary split has already started to resemble a decision tree. This is not a coincidence: making a binary split in a dataset and comparing outcomes in each half is exactly the process at the heart of the decision tree generation algorithm. In fact, [Figure 9-2](#) can rightfully be called a decision tree, albeit one that has only one branching node. We can use [Figure 9-2](#) as a very simple predictor of happiness: we find out how often someone goes out socially. If their `sclmeet` value is 5 or less, then we can predict that their happiness is 7.2. If it is higher than 5, then we can predict that their happiness is 7.8. It will not be a perfect prediction, but it's a start and it's more accurate than random guessing.

We can try to use our decision tree to draw conclusions about the impact of various characteristics and lifestyle choices. For example, we see that the difference between low social happiness and high social happiness is about 0.6, and we conclude that increasing one's level of social activity from low to high could lead to a predicted increase in happiness of about 0.6 on a 10-